Constraint Encoding and Penalty Weight Determine Quantum-Inspired Solver Performance on a Real Crop-Allocation QUBO

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***Abstract*—When many farmers independently choose the same profitable crop, the combined harvest depresses its price. We write this for 174 real parcels (1,464 ha) in Jenin, Palestine as a 1,914-variable QUBO with market-clearing prices, and benchmark two constraint encodings against two solvers over 20 seeds at a matched wall-clock budget. Encoding, not solver, decides usability: under one-hot encoding discrete simulated bifurcation leaves 130 of 174 parcels invalid and earns 67% less than annealing, while domain-wall encoding raises its profit by 138% and makes the two solvers indistinguishable. We then sweep the domain-wall ordering penalty, a weight that published comparisons fix without reporting, and find it inverts the apparent conclusion: annealing loses 22% at the conventional weight and gains 18% at twice that value. A myopic greedy plan loses $10.5M by flooding the market. Both learned components are reported as negative results.**

***Index Terms*—quantum-inspired optimisation, simulated bifurcation, QUBO, domain-wall encoding, constraint encoding, crop planning**

I. Introduction

Crop planning is an allocation problem constrained by a market signal no individual farmer can observe. A farmer plants on last season's prices without knowing what the neighbours intend; when many independently converge on the same profitable crop, the combined harvest depresses its price. Every decision is individually rational and the aggregate outcome is a collective loss, a coordination failure that bites hardest where output cannot be held back until prices recover. Producer prices in Palestine move sharply year to year: across 2010–2023 the largest single-year change in the FAOSTAT series [1] ranges from 14% for wheat to 60% for oranges, and the two crops in our set that store well, wheat and olives, sit at the bottom of that range at 14% and 23%. Recent crop-planning models therefore treat market absorption as a binding constraint rather than assuming prices are exogenous to the plan [2], [3].

Allocating P parcels among C crops admits C^P candidate assignments and does not decompose: because revenue depends on aggregate production, the best crop for one parcel depends on the choices made for all the others. Problems of this form map onto quadratic unconstrained binary optimisation (QUBO) [4], [5], the input format shared by quantum annealers and by a growing family of quantum-inspired classical solvers. Among the latter, simulated bifurcation (SB) [6] integrates the classical Hamiltonian dynamics of nonlinear oscillators. All variables update simultaneously, so SB parallelises where sequential methods such as simulated annealing [7] cannot, and it has cut time-to-solution substantially against annealing hardware on large benchmarks [8].

Passing a discrete allocation problem to any such solver first requires an *encoding*: a representation of each parcel's crop choice in binary variables, together with a mechanism enforcing that exactly one crop is selected. One-hot encoding imposes this through a quadratic penalty over all pairs of binaries within a parcel. Domain-wall encoding [9] instead constrains a chain of binaries to be non-increasing, replacing the dense per-parcel penalty with a local penalty between adjacent binaries. Encoding is ordinarily treated as an implementation detail, subordinate to the choice of solver. On the instance studied here it is not: it decides whether the quantum-inspired solver returns a usable plan at all.

A second decision is buried one level deeper. The domain-wall ordering penalty carries a weight that published comparisons fix once and do not report. We show that this weight determines not only the size of the encoding effect but its *sign* for one of the two solvers, so a single-weight study can reach the opposite conclusion from a swept one.

Our contributions are threefold. First, a 2×2 encoding-by-solver benchmark on a real 1,914-variable agricultural QUBO, over 20 seeds at a verified wall-clock budget, in which domain-wall encoding raises discrete simulated bifurcation's profit by 138% and renders the two solvers statistically indistinguishable. Second, a sweep of the domain-wall ordering penalty across five weights showing that the conventional value sits in a region where annealing appears to lose 22%, while at twice that weight domain-wall improves both solvers; the apparent conclusion inverts with an unreported hyperparameter. Third, an evaluation of every learned component against a trivial baseline, in which an LSTM price forecaster fails to beat persistence and a Random Forest suitability model is shown to carry almost no parcel-level discriminating power, both reported rather than omitted.

II. Literature Review

A. AI-Based Crop Recommendation and Price Forecasting

Random Forest, an ensemble of bagged decision trees, remains a strong baseline for tabular agricultural data because it absorbs nonlinear, heterogeneous features without extensive preprocessing [10]. Dey et al. [11] reached classification accuracy above 99% for crop recommendation with XGBoost on

soil and climatic features, but their pipeline stops at suitability: with no forecasting or allocation stage, a crop flagged suitable may still be a poor choice if the market is saturated. Saha et al. [12] combined satellite land mapping, machine-learning crop and yield prediction above 95% accuracy, and irrigation control; their LSTM forecasts vegetation area rather than prices, so it cannot signal oversupply before it occurs.

Because the allocation objective is driven by prices, the quality of the price input matters as much as the optimiser; Li et al. [2] address this by feeding neural-network forecasts into theirs. Long Short-Term Memory networks [13] are widely applied to agricultural price series for the same purpose. Their benefit is sensitive to record length, and a governorate-level history offers only a few decades of annual observations per crop. Whether a learned forecaster improves on a naive baseline at this data scale is an empirical question rather than a given, and we treat it as one.

B. Optimisation: Classical and Quantum-Inspired

Crop allocation was traditionally posed as linear programming, but recent work treats it as a nonlinear problem integrating prediction and market effects. Li et al. [2] combined simulated annealing with a genetic algorithm across 34 plots in northern China, constraining production by regional market-absorption capacity and farm water supply, and beat standalone GA, SA, PSO and ACO by 5–10 percentage points of profit increment. Hou et al. [3] paired particle swarm optimisation with simulated annealing and Monte Carlo simulation, lowering prices when production exceeds demand, and raised expected profit by 12.6% over a baseline PSO.

Quantum-inspired approaches offer a different route. Zeng et al. [8] benchmarked quantum-annealing-inspired algorithms against a D-Wave annealer on Max-Cut instances of up to 20,000 nodes: ballistic SB cut time-to-solution nearly fifty-fold on small and Chimera-structured graphs, and discrete SB was fastest on large ones. On dense QUBO instances, Kim et al. [14] found a hybrid quantum-annealing solver roughly 6,561× faster than the best classical solver. Within agriculture, Verma et al. [15] paired a quantum convolutional network with QAOA and reported speed-ups of up to 3.5×.

Encoding has been studied separately from solvers. Chen et al. [16] compared domain-wall against one-hot on D-Wave annealing hardware across three problems and found domain-wall superior on every metric, with none favouring one-hot. Berwald et al. [17] proved domain-wall optimal in binaries per discrete variable, showed its dynamics freeze later on hardware, and conjectured that the advantage extends to gate-model and quantum-inspired algorithms.

Two gaps follow. First, none of the agricultural studies formulates allocation as a QUBO with market-clearing prices for Ising-type solvers. Second, the encoding comparisons were run on annealing hardware and abstract problems, within a single solver class, at one ordering-penalty weight. Whether the conjecture of Berwald et al. holds for a quantum-inspired solver on a real allocation problem, and whether the comparison is stable in the penalty weight, have not been tested. Table I summarises the studies reviewed.

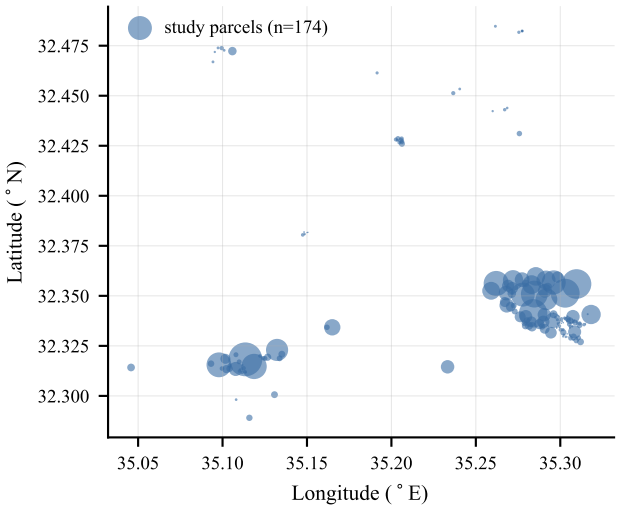


Fig. 1. The 174 study parcels in Areas A/B of Jenin governorate.

III. Methodology

A. Study Area and Data

All parcel, climate, soil, price and yield inputs come from public sources. Parcels are OpenStreetMap land-use polygons [18]. Of 941 polygons, 316 have a centroid in the West Bank; these were assigned an Oslo class from the OCHA layer [19], which does not separate Areas A and B, and restricted to 0.1–100 ha. This leaves 174 A/B parcels (1,464 ha; Fig. 1) and 110 Area C parcels (706 ha).

Monthly NASA POWER data for 2015–2024 [20] give 465 mm of annual rainfall (90% in November–March), 20.7 °C and 68.7% relative humidity. The governorate is a single grid cell, so climate is the same for every parcel. Soil properties are SoilGrids 250 m estimates [21] for 149 A/B parcels; the other 25 take the governorate mean.

Prices are FAOSTAT producer means for 2019–2023 [1]. The year 2024 was left out because perishable prices rose 1.24–5.76-fold over 2023 while wheat (0.95) and olives (1.10) did not, indicating a wartime supply shock. Yields are FAOSTAT means for 2014–2022. National vegetable yields include greenhouse production, so vegetables and maize were scaled jointly to a mean of 19.65 t/ha, the PCBS open-field level for the northern West Bank [22], and tree crops by the factor that brings olives to 1.43 t/ha. Net irrigation needs follow FAO-56 [23]. Production cost is an assumed fraction of baseline gross revenue for each production system, held fixed when prices vary (Table II). The water budget is the 1.0 Mm^3/yr of licensed agricultural abstraction reported by PCBS [24].

B. Suitability Model

A 300-tree Random Forest [10] was trained on N, P, K, temperature, humidity, pH and rainfall. The real data are 1,900 labelled samples over 19 crops from a public dataset [25],

TABLE I
SUMMARY OF REVIEWED STUDIES ACROSS CROP RECOMMENDATION, FORECASTING, AND OPTIMISATION

Study	Focus area	Method	Limitation relative to this study
Breiman [10]	Ensemble learning, general	Random Forest (foundational)	Not agricultural; the method used here for suitability
Hochreiter & Schmidhuber [13]	Sequence modelling, general	LSTM (foundational)	Not agricultural; the method used here for forecasting
Dey et al. [11]	Crop recommendation	XGBoost on soil/climate features	Suitability only; no forecasting or allocation
Saha et al. [12]	Land suitability, irrigation	ML + LSTM (vegetation), satellite data	LSTM forecasts vegetation area, not prices
Li et al. [2]	Dynamic multi-plot crop planning	Hybrid SA–GA + neural-network forecasts	Classical; no QUBO or encoding analysis
Hou et al. [3]	Crop planning under uncertainty	Hybrid PSO–SA + Monte Carlo	No learned forecasts; classical
Zeng et al. [8]	Quantum-inspired optimisation	SB and simulated CIM vs. D-Wave annealer	Abstract benchmarks; solvers, not encodings
Kim et al. [14]	Large-scale dense QUBO	Hybrid quantum annealing vs. classical	Benchmark instances; solvers, not encodings
Verma et al. [15]	Quantum computing in precision agriculture	QCNN + QAOA	No market-clearing prices; no encoding analysis
Chancellor [9]	Encoding discrete variables	Domain-wall encoding	Theoretical; not evaluated on a real problem
Chen et al. [16]	Encoding comparison, experimental	Domain-wall vs. one-hot on D-Wave hardware	Annealing hardware; one solver class, one penalty weight
Berwald et al. [17]	Encoding theory and dynamics	Variable-count optimality; freezing behaviour	Conjectures the extension without testing it

TABLE II
PARAMETERS OF THE TEN STUDY CROPS

Crop	Price (USD/t)	Yield (t/ha)	Net irrigation (m^3/ha)	Cost (% of revenue)
Wheat	568	2.41	500	65
Maize	572	5.79	5000	65
Potatoes	488	13.27	4000	55
Tomatoes	628	51.74	6000	55
Cucumbers	614	18.49	4500	55
Onions	526	9.56	3500	55
Eggplants	459	19.04	6000	55
Olives	1752	1.43	1500	40
Grapes	987	9.38	3000	40
Oranges	601	19.38	7000	40

after its maize, grape and orange rows were removed. A further 1,000 synthetic samples, generated from FAO EcoCrop envelopes [26], label all ten study crops. The two sources are evaluated separately. For crop j on parcel i,

$$s_{ij} = \mathrm{RF}_j(\mathbf{x}_{ij})\,\phi(\ell_i, j), \tag{1}$$

where $\mathbf{x}_{ij}$ holds the parcel's soil and climate with the rainfall feature set to annual rainfall plus the irrigation of crop j, RF_j is the unnormalised class probability, and ϕ weights the OSM class ℓ_i: 0.9 for tree crops and 0.3 for annuals on orchards and vineyards, and 0.9 for annuals and 0.35 for tree crops elsewhere. P and K have no parcel-level source and were drawn from published ranges in 200 Monte Carlo samples, giving s_{ij} a mean standard deviation of 0.025. Available N is an assumed 2% of SoilGrids total N. A pair is unsuitable if its probability is zero or fewer than 20% of A/B parcels score at or below it for crop j; with heavy ties, this marks 96 of 1,740 pairs.

C. QUBO Formulation

The binary x_{ij} equals 1 when parcel i grows crop j, with $j = 0$ for fallow. With parcel area a_i and yield y_j, supply is $Q_j = \sum_i a_i y_j x_{ij}$, and the plan maximises

$$\sum_j \left(p_j Q_j - \beta_j Q_j^2\right) - \sum_{i,j} \left(c_j + \lambda w_j\right) a_i x_{ij}, \tag{2}$$

where p_j is the price, c_j the cost and w_j the net irrigation per hectare, and $\lambda = 0.15\,\mathrm{USD/m^3}$ unless swept. With $\beta_j = p_j/(2D_j)$, price falls linearly with supply and crop revenue peaks at $Q_j = D_j$. The Q_j^2 term couples all parcels sharing a crop into ten 174-node cliques. No crop-level demand data exist for Jenin, so market size is an assumption, $D_j = 0.6\,A_{AB}\,y_j$ with A_{AB} the total A/B area, kept fixed across counterfactuals. For the solvers, the negated objective is divided by its largest linear coefficient, and penalties of weight $M = 0.5$ are added for parcels without exactly one crop and for unsuitable pairs. Profit is recomputed in USD from each decoded plan, without the water charge.

D. Encodings and Solvers

One-hot encoding uses $C = 11$ binaries per parcel with the penalty $M(\sum_j x_{ij} - 1)^2$, giving 1,914 variables; parcels with zero or several crops remain reachable. **Domain-wall** encoding [9] uses $C - 1 = 10$ binaries d_{ij} per parcel, kept non-increasing in j by an ordering penalty of weight M_{wall}, and reads $x_{ij} = d_{ij} - d_{i,j+1}$ with $d_{i0} = 1$ and $d_{iC} = 0$ (1,740 variables). It replaces only the one-crop penalty, so both encodings pose the same problem. Decoding takes a one-hot block's lowest-indexed active option (fallow if none) and reads a domain-wall chain up to its first zero; invalid parcels are

counted before decoding. The main benchmark uses $M_{\text{wall}} = 1$; Section IV also sweeps $M_{\text{wall}} \in \{0.25, 0.5, 1, 2, 4\}$.

SA [7] flips single bits with $O(n)$ incremental energy updates and cools geometrically from $T_0 = 0.5\,\text{mean}_i \sum_k |Q_{ik}|$ to $10^{-4}T_0$. **dSB** [6] evolves positions x_i and momenta y_i as

$$\begin{aligned} \dot{y}_i &= -\,(a_0 - p(t))\,x_i - c_0\Big(\sum_k J_{ik}\,\text{sgn}(x_k) + h_i\Big), \\ \dot{x}_i &= a_0 y_i, \end{aligned} \tag{3}$$

where J and h are the Ising couplings and fields, with inelastic walls at $|x_i| = 1$, $a_0 = 1$, a linearly rising pump $p(t)$, step 0.25 and $c_0 = 0.5/(\sqrt{N}\,\sigma_J)$. Both solvers return the lowest-energy state visited; dSB evaluates its sign state every 50 steps and at the end. **Greedy** gives each parcel its highest-margin suitable crop, ignoring market coupling. The Budget column of Table IV reports profit after fallowing parcels of every plan, greedy included, in ascending margin per m^3 until the water budget is met; other profits precede this repair.

Protocol. Each arm received a fixed wall-clock budget of 2.5 s per run, which corresponds to 2,934 and 4,313 dSB steps and 270 and 249 SA sweeps on the one-hot and domain-wall matrices. Arms whose median time deviated by over 12% were re-run, and results were rejected beyond 15%. Each arm used 20 independent seeds, compared by two-sided Mann–Whitney U tests.

IV. Results

A. Learned Components

In 5-fold stratified cross-validation the Random Forest reaches 0.9958 ± 0.0021 accuracy on the real rows against a majority-class baseline of 0.0526, and 0.8940 ± 0.0297 on the synthetic rows, which measures fidelity to the EcoCrop envelopes rather than predictive skill. Feature importances are rainfall 0.275, humidity 0.174, K 0.150, P 0.134, temperature 0.092, N 0.091 and pH 0.083.

These figures overstate what the model contributes here. Within Jenin, soil pH spans only 7.45–7.80 and available nitrogen 87–138 kg/ha across the 174 parcels, a range narrower than almost any split the forest learned on a global training set. In consequence the 90 distinct soil profiles present in the data collapse to between 2 and 12 distinct suitability values per crop. Parcel-level suitability is therefore close to a per-crop constant modulated by a three-valued land-use factor, and the model should not be read as discriminating between individual parcels.

The optimiser requires a price for each crop. To decide whether that price should be forecast or taken from recent history, we compared an LSTM [13] of 24 hidden units and 3,481 parameters against four naive baselines under walk-forward validation, with the split made before normalisation (Table III).

The LSTM ranks last. With 142 training windows drawn from 13–27 annual observations per crop, 3,481 parameters are not constrained. The optimisation therefore uses 2019–2023 mean prices. Including 2024 in the test window reverses the ranking (LSTM 624.2 against 639.6 for persistence), but the wartime shock raises persistence error 4.9-fold and LSTM error only 2.5-fold, so a window containing 2024 rewards robustness to a discontinuity rather than forecasting skill.

TABLE III
Price forecasts, 50 test windows, 2024 excluded

Model	RMSE ($/t)	MAPE (%)
Persistence	**130.1**	**16.3**
AR(1)	150.5	18.3
Mean of last 3	170.1	19.7
Linear trend	186.0	20.9
LSTM (pooled, augmented)	252.2	26.7

TABLE IV
Encoding by solver, 20 seeds, 2.5 s per run

Enc.	Solver	Profit (M$)	Invalid	Water (Mm^3)	Budget (M$)
one-hot	SA	6.18 ± 0.23	0.0	6.34	1.69 ± 0.20
one-hot	dSB	2.04 ± 0.42	130.3	2.31	1.27 ± 0.29
dom.-wall	SA	4.80 ± 0.53	5.0	7.17	1.75 ± 0.18
dom.-wall	dSB	4.87 ± 0.48	3.5	7.01	1.68 ± 0.18
Greedy		-10.47	—	8.97	1.37

B. Encoding Versus Solver

Table IV and Fig. 2 give the results at $M_{\text{wall}} = 1$. Under one-hot encoding dSB leaves 130 of 174 parcels invalid and earns 67% less than SA ($p < 10^{-7}$): its continuous dynamics bifurcate variables independently, and the dense per-parcel penalty does not reliably leave exactly one positive. Domain-wall encoding raises dSB profit by 138% ($p < 10^{-7}$), after which the two solvers do not differ (-1.5%, $p = 0.78$). At this penalty weight domain-wall *lowers* SA profit by 22% ($p < 10^{-7}$). Section IV-C shows that this last result is an artefact of the weight.

The greedy plan loses $10.47M. Assigning every parcel its best-margin crop concentrates production far beyond D_j; even when unsaleable surplus is discarded rather than sold at the negative prices a linear demand curve implies beyond $2D_j$, it still loses $7.51M. This is the coordination failure the formulation exists to avoid.

C. The Ordering-Penalty Weight

The ordering penalty is a free weight, and published comparisons fix it without reporting a sweep. Table V and Fig. 3 show its effect over $M_{\text{wall}} \in \{0.25, 0.5, 1, 2, 4\}$, 20 seeds per point, against the one-hot references of $6.05M (SA) and $2.04M (dSB) from the same experiment.

Profit rises monotonically with the weight for both solvers until it saturates near $M_{\text{wall}} = 2$. Invalid parcels fall with it, from 23.0 (SA) and 20.3 (dSB) at the weakest weight through 5.0 and 3.4 at $M_{\text{wall}} = 1$ to 0.1 and 0.0 at $M_{\text{wall}} = 2$, where the seed variance also collapses to ± 0.03M: a strong chain penalty does not merely permit valid solutions, it makes the landscape tractable. Domain-wall exceeds one-hot for dSB at every weight tested, by 67% at the weakest and 226% at the strongest. For SA the comparison changes sign: a 45% deficit

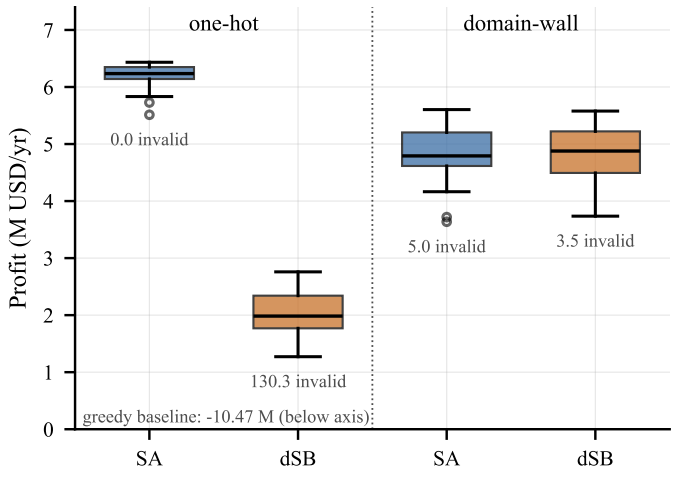


Fig. 2. Profit over 20 seeds, with the mean number of parcels whose binary block lies outside the encoding's valid subspace.

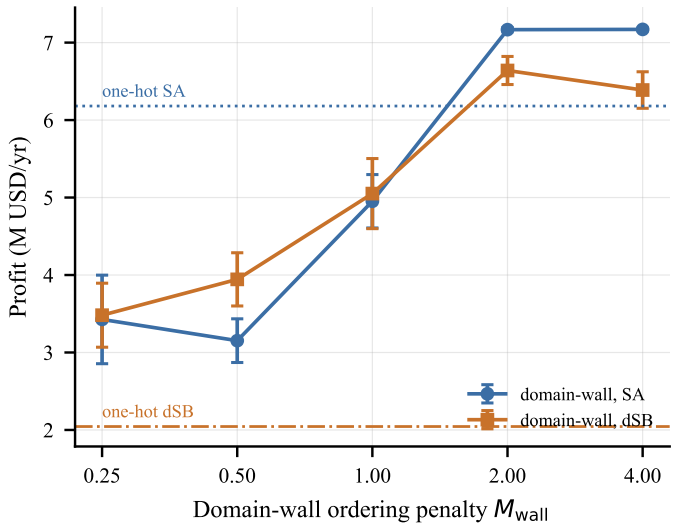


Fig. 3. Profit against the domain-wall ordering penalty, 20 seeds per point, with the one-hot references as horizontal lines. The sign of the annealing effect changes between $M_{\text{wall}} = 1$ and 2.

at $M_{\text{wall}} = 0.25$, 17% at the conventional value of 1, and an 18% *advantage* at 2 and above.

Two conclusions follow. The domain-wall advantage reported on annealing hardware [16] does extend to a quantum-inspired solver on a real allocation problem, confirming the conjecture of Berwald et al. [17], and at an adequate weight it extends to annealing as well. But the conclusion a single-weight study reaches depends on which weight it picked: at the value used in Section IV-B the same experiment reports that domain-wall harms annealing. The weight belongs in the experimental protocol, not in the defaults.

D. Water and the Budget-Feasible Plan

PCBS reports 1.0 Mm^3/yr of agricultural abstraction from licensed Palestinian wells in Jenin governorate, against 5.0 Mm^3 domestic and 6.0 Mm^3 total [24]. Every plan in

TABLE V
DOMAIN-WALL PROFIT AGAINST THE ORDERING PENALTY, 20 SEEDS

M_{wall}	SA (M$)	invalid	dSB (M$)	invalid
0.25	3.43 ± 0.57	23.0	3.48 ± 0.41	20.3
0.50	3.15 ± 0.28	13.6	3.94 ± 0.34	11.1
1.00	4.95 ± 0.34	5.0	5.05 ± 0.45	3.4
2.00	7.17 ± 0.03	0.1	6.64 ± 0.18	0.0
4.00	7.17 ± 0.02	0.0	6.39 ± 0.24	0.0

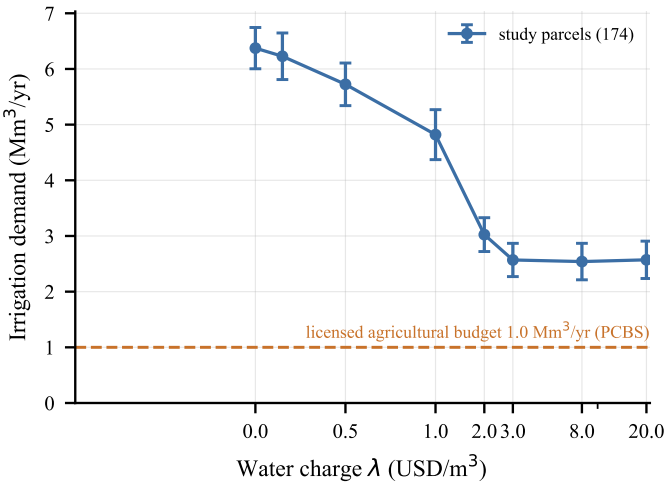


Fig. 4. Irrigation demand against the water charge λ (SA, one-hot, 50 seeds; bars are one standard deviation). Note the symmetric-log axis.

Table IV uses 2.3–9.0 Mm^3, so the licensed allocation binds hard.

Sweeping the water charge (Fig. 4) traces a demand curve that falls monotonically from 6.37 ± 0.37 Mm^3 at $\lambda = 0$ to 2.57 ± 0.30 Mm^3 at \$3/m^3, then flattens: demand never falls below 2.54 Mm^3 at any charge up to \$20/m^3, still 2.5 times the licensed budget. Pricing cannot reach this constraint. Above $\lambda \approx 1$ the formulation is also numerically degenerate, because once the water term dominates the normalising coefficient the profit differences fall below the fixed penalty weights, so only small λ is interpretable.

The budget is therefore imposed directly, by fallowing parcels in ascending order of margin per cubic metre until the plan fits, applied identically to every arm including greedy. The resulting budget-feasible profits are the last column of Table IV. They fall into a narrow \$1.27–1.75M band, cultivating only 9–47 of the 174 parcels, and the encoding difference for SA becomes insignificant ($+3.2\%$, $p = 0.31$) while for dSB it persists ($+32.7\%$, $p < 10^{-4}$). Under the licensed allocation the problem is small enough that encoding and solver choice largely stop mattering. Greedy reaches \$1.37M under the same constraint: forced to leave most land fallow, it can no longer flood the market, so the water limit incidentally performs the coordination that the optimiser performs deliberately.

This deserves stating plainly rather than leaving implicit: at Jenin's licensed allocation the problem is small enough that a trivial heuristic is competitive, and the optimisation

TABLE VI
CROP-MIX SIMILARITY TO THE BASELINE PLAN, 50 SEEDS

Scenario	Profit (M$/yr)	Mix similarity
Seed control	*5.77*	*94.6%*
LSTM prices / persistence	6.05 / 5.82	94.7 / 93.7%
Uniform -15% / $+15\%$	3.63 / 8.06	95.0 / 95.0%
Uniform $+30\%$ / perish. $+30\%$	10.36 / 9.90	93.3 / 94.8%
Uniform -30% / perish. -30%	1.72 / 2.61	87.7 / 82.4%

machinery earns little. It does not follow that the encoding result is contingent on that. The result is a property of a 1,914-variable QUBO with ten 174-node cliques, not of Jenin's water allocation: any allocation problem of comparable size and coupling density presents the same encoding choice, and the water constraint shrinks this instance without shrinking that class. The agricultural case is a testbed that happens to be real, and the two findings bind at different scales, which is why the unconstrained comparison and the budget-feasible plan are reported separately rather than merged.

E. Price Robustness

The optimum is degenerate: inverse demand equalises marginal revenue across crops, so many parcel-level plans reach similar profit and two runs differing only in seed agree on 11.5% of parcels. We therefore compare crop mix by area, as $1-$ total variation distance, against a seed-control floor of 94.6% (50 seeds each).

Every moderate scenario sits at that floor and is therefore indistinguishable from solver noise (Table VI). Only a 30% price fall shifts the crop mix materially, to 87.7% for a uniform fall and 82.4% for perishables alone, moving land toward grapes, olives, oranges and fallow as profit drops to \$1.72M and \$2.61M. The asymmetry is economically sensible: a uniform rise scales revenues together and barely changes relative margins, whereas a fall pushes marginal crops below cost and forces reallocation toward low-cost perennials.

V. LIMITATIONS

Market size D_j is an assumption and the linear demand curve carries no meaning beyond $Q_j = 2D_j$, so every absolute profit reported here inherits that assumption. The encoding and solver comparisons do not: both encodings are built from the same instance, so a different D_j rescales all four arms together and leaves their ordering, and the percentage differences between them, intact. The agricultural magnitudes are a demonstration; the computational comparison does not rest on them. Climate is a single grid cell; the suitability model has almost no parcel-level resolution, for the reasons in Section IV-A; phosphorus and potassium are sampled, not measured; SoilGrids values are model estimates; and yields are national series rescaled to regional open-field levels. The PCBS figure covers licensed wells only, excluding springs, purchased supply, wastewater reuse and unlicensed abstraction, so it is a lower bound and the budget-feasible plans are conservative. Reported profits are of decoded plans, so a solver leaving many blocks outside the valid subspace is scored after an implicit argmax repair. Wall-clock matching is enforced to 15%, not exactly, on a host that showed threefold timing variation. At 1,914 variables with ten 174-node cliques the instance exceeds the largest fully connected problem directly embeddable on current annealing hardware (K_{150} on Pegasus [27]), so this compares classical against quantum-inspired solvers, not quantum hardware.

VI. CONCLUSION

On a real 1,914-variable crop-allocation QUBO, constraint encoding decided whether a quantum-inspired solver was usable at all, and the weight of the domain-wall ordering penalty decided the sign of the effect for annealing. The domain-wall advantage therefore does extend from annealing hardware to quantum-inspired solvers, as Berwald et al. [17] conjectured, but any single-weight comparison risks reporting its opposite. Held to the licensed agricultural water available in Jenin, every configuration converges to \$1.3–1.8M on a sixth of the land, and the choice of encoding and solver largely ceases to matter: the computational question and the agricultural one bind at different scales. Future work should tune the ordering penalty per solver rather than globally, measure freezing behaviour to test the mechanism proposed by Berwald et al., and obtain parcel-level soil and crop-level demand data adequate to the allocation question.

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